Negative Literature Answer to Question 5.3 of arXiv:2105.09203

FA/Banach run packet

2026-06-18

Status

This is a literature-already-answered packet. It records that Question 5.3 from Ansorena–Bello–Wojtaszczyk [1] was answered negatively in the later paper of Albiac–Ansorena–Berasategui [2]. It is not an original proof from this run.

Original Problem

In the open-problems section of [1], the third question, cited later as Question 5.3, asks:

> Let $\mathcal{B}$ be a bidemocratic basis of ℓ_2. Are there $1 < q < r < \infty$ such that
>
> $$\ell_{2,q} \overset{\mathcal{B}}{\hookrightarrow} \ell_2 \overset{\mathcal{B}}{\hookrightarrow} \ell_{2,r}?$$

Supporting Answer

The supporting paper [2] constructs bidemocratic bases with large unconditionality parameters. In Corollary 4.6 it obtains, for separable Hilbert space in particular, a bidemocratic total basis $\mathcal{B}$ with

$$\tilde{k}_m[\mathcal{B}] \approx \log(1+m).$$

The following remark observes that any basis squeezed between suitable Lorentz spaces $d_q(w)$ and $d_r(w)$, with $q > 1$ or $r < \infty$, must satisfy a strictly smaller logarithmic growth estimate for k_m. It then explicitly states that, for Hilbert space, the constructed basis is a counterexample that solves [1, Question 5.3] negatively.

Literature Answer. *Question 5.3 of [1] has a negative answer. There exists a bidemocratic basis of ℓ_2 for which no exponents $1 < q < r < \infty$ give the embeddings $\ell_{2,q} \overset{\mathcal{B}}{\hookrightarrow} \ell_2 \overset{\mathcal{B}}{\hookrightarrow} \ell_{2,r}$.*

Literature identification. The basis supplied by Corollary 4.6 of [2], with the ambient space chosen to be Hilbert space, is bidemocratic and has logarithmic unconditionality growth. The subsequent remark in [2] says that a basis with the Lorentz squeezing requested in Question 5.3 of [1] would have to satisfy

$$k_m[\mathcal{B}] \lesssim (1+\log m)^{1/q-1/r}.$$

For $1 < q < r < \infty$, the exponent $1/q - 1/r$ is strictly less than 1, contradicting the logarithmic lower growth of the constructed basis. The supporting authors record exactly this implication and state that the Hilbert space instance is a negative solution of [1, Question 5.3]. □

Scope Limitations

This packet answers only Question 5.3 of [1]. The preceding questions in that open-problems section, concerning quasi-greedy bases in superreflexive spaces and dual quasi-greediness in ℓ_p, are not claimed here.

Verification Notes

The source and supporting papers are distinct. The supporting paper explicitly cites [1, Question 5.3], so this belongs in `literature_already_answered`. The local duplicate search covered arXiv ids `2105.09203` and `2205.09478`, plus phrases `Question 5.3`, `bidemocratic basis`, `ell_2`, `Lorentz`, and `sparse approximation using new greedy-like bases`.

References

[1] J. L. Ansorena, G. Bello, and P. Wojtaszczyk, *Lorentz spaces and embeddings induced by almost greedy bases in superreflexive Banach spaces*, arXiv:2105.09203.

[2] F. Albiac, J. L. Ansorena, and M. Berasategui, *Sparse approximation using new greedy-like bases in superreflexive spaces*, arXiv:2205.09478.